

SHIVANSH RAO

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EDUCATION

Pennsylvania State University

Master of Science in Computer Vision | **GPA: 4.0/4.0**

Key courses: Deep Learning, Advanced Computer Vision, Natural Language Processing

PA, USA

Dec 2021

Delhi Technological University

Bachelor of Technology in Electronics and Communication Engineering | **GPA: 8.64/10**

Key courses: Machine Learning, Digital Image Processing, Computer Vision, Pattern Recognition, Algorithm Design

Delhi, India

May 2019

WORK EXPERIENCE

Glass Imaging Inc. | Senior ML R&D Engineer / Applied ML Lead | San Mateo, CA

May 2024 - Present

Skills - Computational Imaging · Image Restoration & Registration · Inverse Problems · Neural ISP Design (RAW→JPEG) · On-Device ML

Customer Delivery

- Led the ML development cycle for the company's **first customer product launch** with a major China smartphone OEM — trained the **end-to-end RAW burst pipeline** (registration, restoration, post-processing) for on-device SDR mode, **4x–30x zoom** on the **200 MP main camera** (16-in-1 Hex Bayer, ~0.56 μm native binned to 2.24 μm), running on-device end-to-end in **~3.5s**.
- Led the R&D iteration with a major US smartphone OEM on the **50 MP main camera (Quad + Binned modes)**, training both **restoration and high-zoom super-resolution** models across **3 zoom modes** for both **SDR and HDR** within on-device runtime budgets of **1.2s (SDR) / 1.6s (HDR)**, secured the customer's image-quality (IQ) sign-off, strengthening the partnership ahead of follow-on engagements.
- Acted as the **technical ML POC across customer programs** — converting OEM image-quality feedback into concrete architecture, loss, and pipeline changes and collaborating with Deep-Optics, Data, & deployment to ship the product.

Core R&D Algorithms

- Architected a **device-specific degradation-modeling pipeline** for RAW-domain super-resolution — modeling each module's **optics (PSF) and sensor noise** to enable optics aware training — cutting per-device training timelines from earlier **~1–2 months** to now **~1–2 weeks** and establishing the foundation for scaling ML across new devices.
- Proposed and trained a new efficient restoration model — **designing custom architectures and loss formulations** — delivering SOTA image quality at low computational complexity within mobile latency/memory/SoC budgets; now the base architecture **reused across multiple device projects**, improving resolution and perceptual quality over the benchmark ISP (SSIM, LPIPS, internal IQ).
- Designed and built a **self-supervised interest-point-based image registration algorithm** for handheld burst capture — now the benchmark for the current registration stack running at **~4 ms per frame** on Snapdragon 8s Gen4.

Qualcomm Technologies Inc. | Senior ML Engineer – XR R&D | San Diego, CA

June 2021 - May 2024

Skills - Image Registration · Visual SLAM · 3D Reconstruction · Hardware-Accelerated CV IP

- Designed and implemented the core **hardware-acceleration IP for scale-invariant feature detection**, rotation-invariant descriptors, edge detection, and matching — the classical-CV foundation for visual SLAM, shipped across Automotive, Smartphone-Camera, and AR/VR business units.
- Optimized the **detector/descriptor IP** to extract **2k features per camera at 10 ms latency** on Snapdragon 8s Gen4, supporting 6-DoF pose estimation (XR), motion estimation (mobile), and the SLAM pipeline (automotive).
- Designed and implemented **3D reconstruction (3DR) IP** generating scene meshes via TSDF integration from depth frames, using a custom 3D sparse-volume grid for surface representation — optimized to **15 fps TSDF integration and 1 fps mesh extraction**, shipped for AR/VR and automotive on Snapdragon XR2 Gen 2.
- Built a learned post-processing method for robust feature matching, optimized for on-device NSP/DSP execution.

PUBLICATIONS & PATENTS

The Need for Neural ISP in the Small-Pixel Era [[Arxiv Preprint](#)]

Summer'26

Glass Imaging Inc.

- Shows that as telephoto pixel pitches shrink toward sub-0.5 μm , optics hit a "physics wall" of geometric aberrations, and a Neural ISP trained on the underlying degradations inverts what stage-wise traditional ISPs cannot — turning small-pixel designs into a net resolution advantage.

General Room Layout Estimation [[Paper](#)]

Fall'20

Advisor: Dr. Lee Giles, Dr. Daniel Kifer, Machine Learning Lab | OmniCV Workshop, **CVPR 2021** (Poster)

- Developed SOTA model for reconstructing Manhattan World 3D room layouts from a single RGB panorama.
- Proposed a computationally efficient network that encodes the whole-room layout of the input panorama as three 1D vectors encoding the positions of floor-wall, ceiling-wall, and wall-wall boundaries.

Emotion Guided Continuous Affect Synthesis [[Paper](#)]

Spring'20

Advisor: Dr. James Wang, Affective Computing Lab | HuMA Workshop, **ACM MM 2020** (Poster)

- Improved the SOTA performance by 2% for the task of synthesizing continuous facial expressions from a static image.
- Developed a StarGAN based network to generate continuous facial emotions based on 2D emotion representation (valence and arousal). Proposed to utilize categorical emotions (e.g., happy, sad) to guide the network for smoother transitions in between synthesized images.

Self Supervised Training for Facial Expression Recognition [[Paper](#)]

Spring'20

Advisor: Dr. James Wang, Affective Computing Lab | BEEU Workshop, **ECCV 2020** (Poster)

- Developed SOTA network with an improvement of 1% for the task of facial expression recognition from videos in the wild.
- Adopted the Student-Teacher learning paradigm that uses a combination of labeled dataset and unlabelled dataset for self-training and trained the model with a multi-level attention mechanism.

Named inventor on 7 published U.S. patent applications:

- **Refining Image Features and/or Descriptors** — [US 20250182460 A1](#) (Published: Jun 2025)
- **Video Rectification for Frames Having Camera Introduced Distortion** — [US 20250292370 A1](#) (Published: Sep 2025)
- **Spatio-Temporal Cooperative Learning for Multi-Sensor Fusion** — [US 20250094535 A1](#) (Published: Mar 2025)
- **Egocentric Gaze Estimation** — [US 20250348135 A1](#) (Published: Nov 2025)
- **Security Systems for Machine Learning Models** — [US 20240400079 A1](#) (Published: Dec 2024)
- **Camera Soiling Detection Using Attention-Guided Camera Depth and LiDAR Range Consistency Gating** — [US 20250085407 A1](#) (Published: Mar 2025)
- **Cache Memory Architecture Augmentation for 3-Dimensional (3D) Data** — [US 20250117876 A1](#) (Published: Apr 2025)

NEWS AND MEDIA

- Invited speaker at **"A Tour Through AI-Powered Photography and Imaging"** lecture series at **International Conference on Computer Vision 2025**, presenting alongside researchers from **Sony AI** and **University of Würzburg**. [[ICCV 2025](#)].
- Invited keynote speaker to inaugurate the seminar series at the **IEEE Computer Society San Diego Lecture Series (2026)**. [[IEEE SD 2026](#)]
- Program Committee Member & Advisor, [AIGENS Workshop](#), CVPR 2026.
- Quoted as the ML POC for GlassAI's Neural ISP in PetaPixel article, explaining the optics modeling and end-to-end RAW pipeline behind the product. [[PUBLISHED ARTICLE](#)]

SKILLS

Programming	Python, C/C++, MATLAB
ML Frameworks	PyTorch, TensorFlow
Computational Imaging	Neural ISP, HDR fusion, RAW burst super-resolution, RAW video super-resolution, Image Registration, Agentic Image-Restoration Pipeline
Deployment	CoreML, CUDA, Distillation, Quantization